BAT Engineering

Hose Management System (HMS) 2.0

Notifications and Alerting Specification

Email and SMS notifications across the HMS 2.0 lifecycles

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1 Document control

Version	Date	Author	Summary of change
0.1	Jun 2026	Architecture	Initial System Design (draft, markdown)
0.2	Jun 2026	Architecture	Mobile & Offline Addendum A (draft, markdown)
1.0	27 June 2026	Architecture	Consolidated Notifications and Alerting Specification; review refinements incorporated

Supersession. This document consolidates and supersedes the earlier System Design and Mobile/Offline Addendum drafts. The field application described in those drafts as an offline *PWA* is superseded: the field client is delivered as a **packaged mobile app** built with Ionic + Angular + Capacitor (Angular running in a native shell), as defined herein. No competing approaches remain.

Contents

2 Introduction

2.1 Purpose

This specification defines email and SMS notifications across HMS 2.0 - the events that trigger them, the recipients, the channel and criticality of each, the supporting subsystem, and testable requirements. It extends the Technical Design Document (asynchronous processing and the Notification entity) and the Functional Specification Document (retest reminders and related capabilities), and is read alongside both.

2.2 Scope

In scope: business notifications to customers, inspectors, reviewers and administrators by email and SMS, plus the in-app/push messages that accompany them. Operational alerting to the engineering team (job and delivery failures) is summarised in section 10 but is delivered through the observability stack rather than the customer notification channels.

2.3 Principles

- **Event-driven.** Notifications fire from committed domain events via the transactional outbox, so a rolled-back change never notifies and every send is retryable.
- **Scheduled where time-based.** A daily scheduler raises advance, due, overdue and approaching-condemnation notifications.
- **Criticality-tiered channel policy.** Safety, security and transactional messages send immediately on both channels and are not opt-out; informational messages are email-first and batchable.
- **Consent and compliance.** Commercial messages obtain consent, identify BAT, and offer unsubscribe; phone numbers are verified before SMS.
- **Tracked and audited.** Every notification is persisted with delivery status and recorded against the asset/customer as compliance evidence.
- **De-duplicated.** An idempotency key per event, recipient and channel prevents duplicate sends.

3 Notification architecture

Two sources feed a dispatcher, which resolves recipients and preferences, renders templates, and routes to channel adapters; delivery is tracked and retried.

3.1 Sources

Domain events (for example certificate issued, inspection rejected, asset condemned, device registered) are written to the transactional outbox in the same transaction as the business change. The **scheduler** (Celery beat) runs daily, queries due/overdue assets and approaching dates, and enqueues the corresponding notification events.

3.2 Dispatcher

A Celery worker consumes notification events and, for each: resolves the recipient set from the event and role mapping; applies the recipient's per-category channel preferences (subject to the criticality override); selects and renders the channel-specific template; and enqueues channel sends. The dispatcher is idempotent on the event/recipient/channel key.

3.3 Channels

Email is sent through OCI Email Delivery; SMS through Twilio. A channel-adapter abstraction isolates providers so either can be swapped or supplemented (for example an alphanumeric sender ID for Australian SMS) without touching dispatch logic.

3.4 Delivery tracking, retry and dead-letter

Each send persists a Notification row whose status is updated by provider webhooks (sent, delivered, failed, bounced). Failures retry with exponential backoff; after a configured number of attempts the message is dead-lettered and an operational alert is raised. Bounced or repeatedly failing addresses/numbers are suppressed and flagged for correction.

3.5 Idempotency and de-duplication

Every notification carries an idempotency key derived from the triggering event, the recipient and the channel, so retried events or overlapping scheduler runs never produce duplicate messages.

4 Criticality tiers and channel policy

Tier	Channels and behaviour
Critical (safety / security)	Email + SMS, immediate, non-batchable, sender-identified. Treated as essential safety/operational messages: not subject to marketing opt-out, and may bypass quiet hours. Examples: asset condemned, inspection failed, certificate revoked, new device registered, suspicious login.
Transactional	Account and identity actions delivered immediately on the appropriate channel (one-time codes by SMS, resets and invitations by email). Factual/transactional content, sender-identified, not marketing opt-out.
Important	Email always; SMS if the recipient has opted in. Prompt delivery; quiet hours respected unless escalating. Examples: retest due and overdue, inspection rejected, certificate issued, inspection awaiting review.
Informational	Email; may be batched into a digest. Respects quiet hours and preferences and is easily opted out. Examples: advance retest reminders, approaching condemnation, role changed, job completion.

5 Recipients and roles

Customer-facing notifications go to the customer's nominated notification contacts; internal notifications go to the relevant BAT role - the account owner for customer/asset events, the assigned reviewer for inspection submissions, the initiating administrator for job completions, and a manager on escalation. Recipient resolution is driven by the event type and role mapping, not hard-coded addresses.

6 Consent, preferences and compliance

Under the Australian Spam Act 2003, commercial electronic messages require consent, clear sender identification, and a functional unsubscribe. HMS 2.0 obtains consent at onboarding, records it, identifies BAT as sender, and provides per-category opt-out for non-essential notifications. Safety, security and transactional messages are treated as essential (factual, not marketing) and are not opt-out, but remain sender-identified. Email addresses and phone numbers are verified before use, and opt-outs are honoured promptly. Recipients manage their preferences in a notification preference centre in the portal/app. (Confirm the safety-message and transactional exemptions with BAT's legal/compliance function; this is design guidance, not legal advice.)

7 Lifecycle notification matrix

The following table catalogues every notification-bearing event across the system's lifecycles.

Lifecycle	Event and trigger	Recipients	Channels	Tier
Retest scheduling	Advance reminder (e.g. 30 and 14 days before due)	Customer contacts; BAT account owner	Email; SMS if opted in	Important

Lifecycle	Event and trigger	Recipients	Channels	Tier
Retest scheduling	Retest due (on the due date)	Customer; BAT account owner	Email + SMS	Important
Retest scheduling	Overdue, with escalations (+7, +14, +30 days)	Customer; BAT account owner; escalate to BAT manager	Email + SMS	Important / Critical
Retest scheduling	Service booked or confirmed	Customer	Email	Informational
Asset	Approaching condemnation (60 and 30 days before grave date)	Customer; BAT account owner	Email	Informational
Asset	Condemned / graved (state becomes CONDEMNED)	Customer; BAT account owner	Email + SMS	Critical (safety)
Asset	Retired	BAT account owner	Email	Informational
Asset	Bulk import completed	Admin who initiated	Email + in-app	Informational
Inspection	Submitted - awaiting review	Assigned reviewer	Email + in-app	Important
Inspection	Approved	Inspector	Email + in-app	Informational
Inspection	Rejected - action needed (re-inspect)	Inspector	Email + SMS	Important
Inspection	Failed result - asset failed inspection	Customer; BAT account owner; reviewer	Email + SMS	Critical (safety)
Certificate	Issued and available	Customer; BAT account owner	Email (with link); SMS if opted in	Important
Certificate	Bulk generation completed	Admin who initiated	Email + in-app	Informational
Certificate	Revoked or superseded	Customer; BAT account owner	Email + SMS	Critical
User / identity	Invitation / account created	Invited user	Email (activation link)	Transactional
User / identity	Password reset requested	User	Email (reset link)	Transactional
User / identity	MFA / one-time code	User	SMS	Transactional (immediate)
User / identity	Suspicious login or account locked	User	Email + SMS	Critical (security)
User / identity	Role or permission changed	Affected user	Email	Informational
Device / mobile security	New device registered for a user	Device owner	Email + SMS	Critical (security)
Device / mobile security	Device revoked by admin	Device owner (device wipes on next sync)	Email + in-app	Critical (security)
Device / mobile security	Offline window expiring - not synced	Device owner	In-app / push; Email optional	Important
Device / mobile security	Repeated sync failures	Operations / admin	Email (operational alert)	Operational

Table 1 - Lifecycle notification matrix. Critical and transactional messages are sent on both channels regardless of marketing preferences (with sender identification); informational messages are email-first and may be batched.

8 Templates

Templates are versioned and channel-specific. Email templates provide both an HTML and a plain-text part, BAT-branded, with a clear call to action and (for non-essential categories) an unsubscribe link. SMS templates are short and include a deep link rather than detail. Templates support localisation and are stored as data so wording can change without a deployment.

9 Data model additions

Entity	Purpose / key fields
NotificationTemplate	key, channel (email/sms), version, locale, subject, body, active
Notification (extended)	event_ref, category, recipient_type, recipient_id, channel, template_key, status, provider_message_id, attempts, idempotency_key, scheduled_for, sent_at, delivered_at, error
NotificationPreference	party_id, category, channel, opted_in, updated_at
Contact verification	email_verified, phone_verified and a normalised E.164 phone on User and CustomerContact

10 Functional requirements and acceptance criteria

ID	Requirement	Acceptance criteria	Phase
N-01	Event-driven dispatch	A notification is sent only after the triggering transaction commits; a rolled-back change produces no notification	2/3
N-02	Scheduled reminders	A daily run enqueues advance, due and overdue retest reminders and approaching-condemnation notices	3
N-03	Dual channel	Email is delivered via OCI Email Delivery and SMS via Twilio through a channel-adapter abstraction	3
N-04	Preferences and opt-in	Recipients manage per-category channel preferences; SMS requires explicit opt-in and a verified phone number	3
N-05	Mandatory safety/security	Critical safety and security notifications are sent regardless of marketing preferences and identify BAT as sender	3
N-06	Delivery tracking and retry	Each notification persists a status updated by provider webhooks; failures retry with backoff and dead-letter after N attempts	3
N-07	De-duplication	The same event never produces a duplicate send to the same recipient and channel	3
N-08	Quiet hours and batching	Non-critical notifications respect recipient quiet hours and may be combined into a digest	4
N-09	Audit	Every notification is recorded (recipient, channel, event, timestamp, status) and is retrievable against the asset/customer	2/3
N-10	Compliance	Non-essential messages identify the sender and provide unsubscribe; opt-outs are honoured promptly	3
N-11	Escalation	Overdue retests escalate through defined steps and recipients up to a BAT manager	3
N-12	Transactional immediacy	Password-reset links and one-time codes are delivered immediately on the appropriate channel	1/3

11 Operational alerting

Distinct from customer notifications, operational alerts cover background-job failures (certificate generation, reminder runs), notification delivery failures, and repeated device sync errors. These are routed to the engineering team through the observability stack (Prometheus/Alertmanager and Grafana) and an operations email or chat channel, are never subject to recipient opt-out, and do not use the customer-facing templates.

12 Open questions

- Which categories are SMS-eligible by default versus email-only, given SMS cost and intrusiveness?
- Are safety notifications strictly mandatory (no opt-out) - to be confirmed with legal under the Spam Act?
- SMS sender configuration for Australia (Twilio number or alphanumeric sender ID) and monthly budget caps?
- Quiet-hours policy and recipient time zones across Australia?
- Exact reminder cadence (days before due) and escalation steps?
- When many of a customer's assets are due at once, per-asset messages or a single digest?